Dear Quillstone Payments Hiring Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. I am a data scientist with six years in analytics and data science, and I am moving towards machine learning engineering. At Thistlebox, I build forecasting and churn workflows used by operations and retention teams, and I design experiments with product and growth colleagues. I am interested in applying this experience to AI systems that help merchants get support quickly and without friction.

My recent work includes a six-week-ahead demand forecasting workflow using LightGBM and engineered calendar, price and menu-position features. Run as a scheduled notebook job, it writes forecasts to BigQuery for operations teams; WAPE fell from 23% to 14%, with an estimated 9% reduction in food waste. I also developed a weekly-scored XGBoost churn model and built a sequential-testing calculator in Python and Streamlit, writing the app and its Dockerfile while the data platform team handled Cloud Run deployment. These projects have given me practical experience with feature engineering, data workflows and collaboration across teams.

I work with Python, BigQuery and dbt, and in 2025 began adding pytest unit tests to forecasting feature code, reaching approximately 60% coverage. My production experience is primarily with scheduled and batch workflows rather than model APIs, and my PyTorch experience is from a learning project. I would bring a strong statistical foundation, practical machine-learning experience and a deliberate focus on developing my production engineering practices. Quillstone's emphasis on ownership, teamwork and learning from mistakes is particularly appealing to me.

Thank you for considering my application. I would welcome the opportunity to discuss how my experience could contribute to the Customer Operations AI team.

Kind regards,
Tobias Mwangi-Hartley